Lopcombe Corner — Peak Level vs Winter Rainfall (Oct–Mar, corrected)

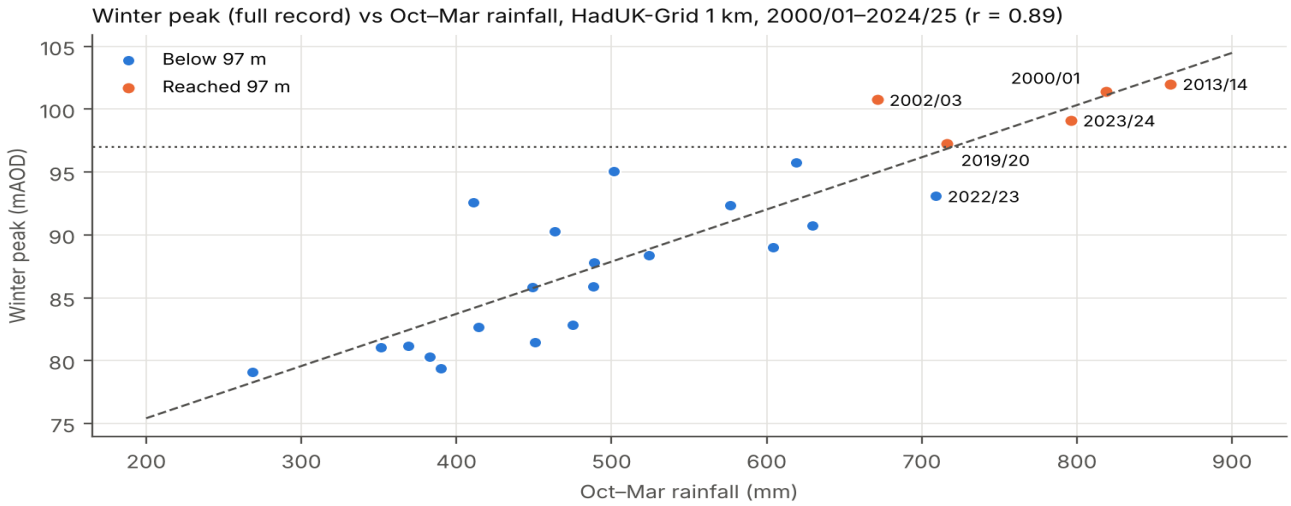
25 winters 2000/01–2024/25 · September statistics: corrected peaks and Broughton rainfall re-summed by month from Broughton E13580 and validated against Tim Vaughan's Broughton figures · analysis 7 Sep 2026

Recreated and updated 20 Oct 2026. Statistics in sections 1–3 are those from the September analysis, which used Broughton rainfall and winter peaks that were later found to be too low in several years. The chart and data table use **HadUK-Grid 1 km rainfall** and winter peaks from the **full daily EA record**. On that basis peak vs Oct–Mar is r = 0.87 for the logger era and r = 0.83 over 62 winters back to 1963/64.

Headline — Oct–Mar is the right window; antecedent low matters too

Window	r	r^2	+ antecedent low
Oct–Dec	0.72	0.52	—
Oct–Jan	0.80	0.64	R^2 0.73
Oct–Mar	**0.86**	**0.74**	**R^2 0.83**

- Extending the window from Oct to Mar lifts r from 0.80 to 0.86: the recharge that sets the peak runs into Feb–Mar.
- Adding the antecedent recession low lifts Oct–Mar R^2 to 0.83 — about +5 m of peak per +1 m of antecedent level — so the buffer effect is a clear contributor once the rainfall window is right.
- Oct–Mar fit (Broughton): peak ≈ 0.047 × rain + 64.4, about +4.7 m per +100 mm.
- Why Oct–Jan fell short: most peaks occur Feb–Mar, fed by late-winter rain — Feb 2020 (148 mm) drove the 24 Feb 2020 peak of 97.25 m; Feb 2014 (141 mm) the 13 Feb record of 102.0 m.



97 m Pitton threshold (revised)

Five of 25 logger-era winters reached 97 m (about 1 in 5): 2000/01, 2002/03, 2013/14, 2019/20 and 2023/24. On the Broughton sums all had Oct–Mar ≥ ~615 mm and none below ~565 mm reached it; on HadUK-Grid all had ≥ ~670 mm. An Oct–Mar running total is a better early indicator than Oct–Jan, but timing and antecedent state still shift the exact peak — 2022/23 was very wet yet peaked at only 93.1 m. A sixth winter, 2025/26 (99.02 m), is not in this table because HadUK-Grid rainfall currently ends in December 2025.

Validation and caveats

- Broughton monthly Oct–Jan sums vs Tim Vaughan: 2001 529 vs 534, 2003 555 vs 557 (~1%), but 2014 565 vs 605, 2020 445 vs 484, 2024 456 vs 509 (7–11% low) — the fetch under-added the wettest months. HadUK-Grid gives 578, 603, 634, 496 and 505 mm.
- Winter 2018/19 Broughton totals under-read (12-day data gap in January 2019).
- Broughton is about 10 km from the borehole; HadUK-Grid uses the borehole's own 1 km cell.

Data: winter peaks (full EA record) and HadUK-Grid 1 km rainfall

Winter	Oct–Dec mm	Oct–Jan mm	Oct–Mar mm	Peak mAOD	Peak date
2000/01	**480**	**578**	**819**	**101.39**	**17 Dec 2000**
2001/02	229	328	488	85.90	3 Apr 2002
2002/03	**511**	**604**	**671**	**100.75**	**8 Jan 2003**
2003/04	278	394	475	82.85	5 Mar 2004
2004/05	263	312	382	80.32	13 Feb 2005
2005/06	245	272	390	79.37	1 May 2006
2006/07	373	466	630	90.76	19 Mar 2007
2007/08	194	311	449	85.85	9 Feb 2008
2008/09	198	292	411	92.60	23 Feb 2009
2009/10	387	457	604	89.00	20 Apr 2010
2010/11	170	272	351	81.06	25 Mar 2011
2011/12	174	222	268	79.08	31 May 2012
2012/13	398	496	619	95.73	11 Feb 2013
2013/14	**402**	**634**	**860**	**102.01**	**13 Feb 2014**
2014/15	312	412	489	87.78	13 Feb 2015
2015/16	221	359	524	88.39	4 Mar 2016
2016/17	159	247	369	81.18	9 Apr 2017
2017/18	210	295	464	90.31	25 Apr 2018
2018/19	268	305	451	81.46	24 Apr 2019
2019/20	**380**	**496**	**716**	**97.25**	**25 Feb 2020**
2020/21	384	474	576	92.33	20 Feb 2021
2021/22	247	272	414	82.68	7 Apr 2022
2022/23	429	559	709	93.09	26 Jan 2023
2023/24	**420**	**505**	**796**	**99.11**	**12 Apr 2024**
2024/25	270	415	501	95.04	11 Feb 2025

Bold rows reached the 97 m threshold. Rainfall: Met Office HadUK-Grid v1.3.2.ceda (release v20260512), cell centred E 425500 / N 135500. Peaks: highest daily logger reading October–May.